

EC400 Probability and Statistical Inference

Pre-Arrival Quiz

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September 2026

Mark all true statements; **more than one may be correct**. All answers will be posted once EC400 begins.

Questions 1–5 correspond to Lecture 1: Foundations.

1. Mark all answers which are true about populations, samples and statistical targets.
 - (a) A population contains all the units about which we want to learn; a sample contains the units we observe.
 - (b) The population defines the target, while the sample is used to estimate it.
 - (c) A population target θ is fixed across repeated samples generated from the same population and design.
 - (d) The same estimate must be obtained from every possible sample of a given size.
2. Suppose $Y = DY(1) + (1 - D)Y(0)$, $P(D = 1) \in (0, 1)$ and $D \perp\!\!\!\perp (Y(1), Y(0))$
 - (a) The average treatment effect is $ATE = E[Y(1) - Y(0)]$.
 - (b) $E(Y \mid D = 1) - E(Y \mid D = 0) = ATE$.
 - (c) For every individual in the experiment, both $Y(1)$ and $Y(0)$ are observed.
 - (d) A difference in means can be causal under one design and merely descriptive under another.
3. Consider repeated sampling from the same population under the same design
 - (a) A sampling distribution describes an estimator across repeated samples or random assignments.
 - (b) A standard error is the sampling standard deviation of an estimator, or an estimate of it.
 - (c) After the sample is observed, the realized estimate is a fixed number.
 - (d) An estimate is the rule $g(\cdot)$ rather than the number produced in the observed sample.
4. Let $\hat{\theta}_n$ be an estimator of a scalar target θ
 - (a) $\hat{\theta}_n$ is unbiased if $E(\hat{\theta}_n) = \theta$.
 - (b) $\hat{\theta}_n$ is consistent if, for every $\varepsilon > 0$, $P(|\hat{\theta}_n - \theta| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.
 - (c) Every unbiased estimator is consistent.
 - (d) Consistency requires $\hat{\theta}_n = \theta$ in every finite sample.
5. Assume the relevant second moments exist
 - (a) $MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2$.

- (b) If $\hat{\theta}$ is unbiased, then $MSE(\hat{\theta}) = Var(\hat{\theta})$.
- (c) An estimator with lower variance necessarily has lower MSE, regardless of its bias.
- (d) A biased estimator can have lower MSE than an unbiased estimator.

Questions 6–10 correspond to Lecture 2: Statistics I.

6. Let X be a continuous random variable with cdf F and density f
 - (a) $F(x) = P(X \leq x)$.
 - (b) $f(x) = P(X = x)$.
 - (c) For $a < b$, $P(a < X \leq b) = F(b) - F(a) = \int_a^b f(t) dt$.
 - (d) A cdf is non-decreasing, with limits 0 and 1 at the left and right tails respectively.
7. Let X have finite variance and let a and b be constants
 - (a) $E(aX + b) = aE(X) + b$.
 - (b) $Var(aX + b) = a^2 Var(X)$.
 - (c) $SD(aX + b) = a SD(X)$ for every real value of a .
 - (d) $Var(X) = E(X^2) - [E(X)]^2$.
8. Let X and Y have joint density $f_{XY}(x, y)$, and suppose $f_Y(y) > 0$ at the value under consideration
 - (a) $f_X(x) = \int f_{XY}(x, y) dy$.
 - (b) $f_{X|Y}(x | y) = \frac{f_{XY}(x, y)}{f_Y(y)}$.
 - (c) $f_{XY}(x, y) = f_X(x)f_Y(y)$ for every pair of jointly distributed random variables.
 - (d) If X and Y are independent, then $f_{X|Y}(x | y) = f_X(x)$ wherever the conditional density is defined.
9. Assume all expectations and variances below exist
 - (a) $E(X) = E[E(X | Y)]$.
 - (b) $Var(X) = E[Var(X | Y)] + Var(E[X | Y])$.
 - (c) If X and Y are independent, then $E(X | Y) = E(X)$ almost surely.
 - (d) If $E(X | Y) = 0$ almost surely, then X and Y must be independent.
10. Consider the distinctions among random sampling, random assignment and iid observations
 - (a) Random assignment can identify a causal treatment comparison within the study sample.
 - (b) Random sampling helps connect a study sample to a target population.
 - (c) An iid sample consists of observations that are mutually independent and have the same marginal distribution.
 - (d) Random assignment and random sampling are two names for the same requirement.

Questions 11–15 correspond to Lecture 3: Statistics II.

11. Let $Z, Z_1, \dots, Z_\nu$ be independent $N(0, 1)$ random variables and let $U \sim \chi_\nu^2$ be independent of Z
 - (a) $Z^2 \sim \chi_1^2$.
 - (b) $\sum_{j=1}^\nu Z_j^2 \sim \chi_\nu^2$.

- (c) $\frac{Z}{\sqrt{U/\nu}} \sim t_\nu$.
- (d) If independent $U_j \sim \chi_{\nu_j}^2$, then $U_1/U_2 \sim F_{\nu_1, \nu_2}$.
12. Mark all answers which are true about the existence of moments.
- (a) If $E(|X|^4) < \infty$, then $E(|X|^2) < \infty$.
- (b) If $E(|X|) < \infty$, then $E(X^2) < \infty$.
- (c) Every probability distribution has finite moments of every positive order.
- (d) If a right-tail density behaves like $f(x) \asymp c/x^2$ for large x , then its second moment is infinite.
13. Assume the expectations below exist
- (a) If g is convex, then $g(E(X)) \leq E[g(X)]$.
- (b) If g is concave, then $g(E(X)) \geq E[g(X)]$.
- (c) Applying Jensen's inequality to $g(x) = x^2$ gives $E(X^2) \geq [E(X)]^2$.
- (d) Equality in Jensen's inequality holds for every convex function and every random variable.
14. Consider the regression notation $y = X\beta + \varepsilon$ with n observations and k regressors, including an intercept if one is used
- (a) y and ε are $n \times 1$ vectors.
- (b) X is $n \times k$ and β is $k \times 1$.
- (c) Row i of X is x'_i , so the scalar equation is $y_i = x'_i\beta + \varepsilon_i$.
- (d) $X'X$ is an $n \times n$ matrix.
15. Let $u = (u_1, \dots, u_n)'$ and let $\Omega = \text{Var}(u)$
- (a) Ω is symmetric and its diagonal entries are the variances $\text{Var}(u_i)$.
- (b) The off-diagonal entry Ω_{ij} is $\text{Cov}(u_i, u_j)$.
- (c) Homoskedastic and mutually uncorrelated errors have $\Omega = \sigma^2 I_n$.
- (d) If every off-diagonal covariance is zero, the components of u must be mutually independent.

Questions 16–20 correspond to Lecture 4: Hypothesis Testing.

16. Consider a hypothesis test of $H_0 : \beta = \beta_0$
- (a) The null hypothesis is the default claim whose compatibility with the data is assessed.
- (b) A p -value is the probability, calculated under H_0 , of observing a test statistic at least as extreme as the realized one.
- (c) A p -value is the posterior probability $P(H_0 \text{ is true} \mid \text{data})$.
- (d) The significance level α should be chosen before looking at the estimate.
17. For a standard two-sided level- α test and its corresponding $(1 - \alpha)100\%$ confidence interval, mark all answers which are true.
- (a) The test rejects when the p -value is smaller than α .
- (b) The test rejects $H_0 : \beta = \beta_0$ when β_0 lies outside the corresponding confidence interval.
- (c) Failure to reject H_0 proves that H_0 is true.
- (d) The critical value depends on the reference distribution and the chosen significance level.

18. Suppose $\hat{\beta} = 0.12$, $\widehat{se}(\hat{\beta}) = 0.05$, and we test $H_0 : \beta = 0$ using a large-sample standard normal reference distribution
- (a) The test statistic is $t = (0.12 - 0)/0.05 = 2.40$.
 - (b) At the 5% two-sided level, the test rejects because $|2.40| > 1.96$.
 - (c) The approximate 95% confidence interval is $[0.022, 0.218]$, up to rounding, and it excludes zero.
 - (d) The two-sided p -value is greater than 0.05.
19. Mark all answers which are true about testing errors and power.
- (a) A Type I error is rejection of a true null hypothesis.
 - (b) A Type II error is failure to reject a false null hypothesis.
 - (c) Power at a particular alternative is one minus the probability of a Type II error at that alternative.
 - (d) Lowering α while holding the design, sample size and alternative fixed generally increases power.
20. Compare one-sided and two-sided tests at the same significance level α
- (a) A one-sided test places the full rejection probability α in one tail.
 - (b) A one-sided test can have greater power against alternatives in its pre-specified direction.
 - (c) It is valid to choose the direction of a one-sided test after observing the sign of the estimate.
 - (d) A two-sided alternative such as $H_1 : \beta \neq \beta_0$ treats sufficiently large departures in either direction as evidence against H_0 .

Questions 21–25 correspond to Lecture 5: Asymptotics. Question 25 covers the lecture’s optional, non-examinable extension.

21. Let $\hat{\theta}_n$ be an estimator and let θ be a constant
- (a) Consistency means $\hat{\theta}_n \xrightarrow{p} \theta$.
 - (b) Convergence in probability implies convergence in distribution.
 - (c) Convergence in distribution to the constant θ implies convergence in probability to θ .
 - (d) Convergence in distribution to a non-degenerate random variable is the same thing as consistency for θ .
22. Let $X_1, X_2, \dots$ be iid with $E(|X_i|) < \infty$ and $E(X_i) = \mu$
- (a) The law of large numbers gives $\bar{X}_n \xrightarrow{p} \mu$.
 - (b) The law of large numbers by itself says that $\sqrt{n}(\bar{X}_n - \mu)$ is approximately normal.
 - (c) Dependent data can also satisfy a law of large numbers, but additional conditions are needed.
 - (d) Law-of-large-numbers arguments are a principal route to proving consistency of sample analogs.
23. Let $X_1, X_2, \dots$ be iid with $E(X_i) = \mu$ and $0 < Var(X_i) = \sigma^2 < \infty$
- (a) $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$.
 - (b) $\bar{X}_n$ is exactly normally distributed for every n , whatever the distribution of X_i .

- (c) The standard deviation of $\bar{X}_n$ is of order $n^{-1/2}$.
 - (d) The classical finite-variance central limit theorem need not apply when $\text{Var}(X_i) = \infty$.
24. Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, V)$ with $0 < V < \infty$
- (a) The stated asymptotic normality implies $\hat{\theta}_n \xrightarrow{p} \theta$.
 - (b) For large n , it suggests the approximation $\hat{\theta}_n \approx N(\theta, V/n)$.
 - (c) If $\hat{V} \xrightarrow{p} V$, an approximate 95% confidence interval is $\hat{\theta}_n \pm 1.96\sqrt{\hat{V}/n}$.
 - (d) Asymptotic normality guarantees that the estimand has the intended causal or economic interpretation.
25. *Non-examinable extension.* Mark all answers which are true about continuous mapping and Slutsky's theorem.
- (a) If $X_n \xrightarrow{p} c$ and g is continuous at c , then $g(X_n) \xrightarrow{p} g(c)$.
 - (b) If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$.
 - (c) If $Y_n \xrightarrow{p} 0$, then X_n/Y_n always has a well-behaved limiting distribution.
 - (d) Slutsky's theorem often justifies replacing an unknown scale parameter by a consistent estimator of it.